

API Documentation

Leeds Bus Reliability and Delay Analytics API - endpoints, parameters, request/response formats, authentication, and error codes.

Base URL (local): `http://127.0.0.1:8000`

1. Authentication

- **Public (no auth):** All GET endpoints.
- **Protected:** POST, PUT, and DELETE under `/incidents` require the `X-API-Key` header.

How to authenticate:

Include the header in every write request:

`X-API-Key: your-write-api-key`

The key is configured in `.env` as `WRITE_API_KEY`. If the header is missing or invalid, the API returns `401 Unauthorized`.

2. Error codes

HTTP status	Meaning
200 OK	Successful read or update.
201 Created	Incident created successfully.
204 No Content	Incident deleted successfully (no body).
401 Unauthorized	Missing or invalid <code>X-API-Key</code> on a protected endpoint.
404 Not Found	Resource not found (e.g. route, stop, incident by ID).
422 Unprocessable Entity	Validation error (invalid body or query parameters).

3. Endpoints

3.1 GET `/operators`

Returns all operators imported from BODS.

Parameters: None.

Example response (200):

```
[
  {
    "id": 1,
    "noc": "YKRT",
    "name": "Example Operator",
    "region": "Yorkshire"
  }
]
```

3.2 GET /routes

Returns route records with optional filters.

Query parameters:

Name	Type	Description
operator_id	integer	Filter by operator ID.
q	string	Search in route code/name.
limit	integer (1-100)	Max results (default 20).
offset	integer (≥ 0)	Skip (default 0).

Example request:

GET /routes?operator_id=1&limit=10

Example response (200):

```
[
  {
    "id": 1,
    "operator_id": 1,
    "route_code": "1",
    "route_name": "City Centre - Chapel Allerton",
    "origin": "City Centre",
    "destination": "Chapel Allerton"
  }
]
```

3.3 GET /routes/{route_id}

Returns one route with operator and ordered stops.

Path parameters: route_id (integer).

Example request:

GET /routes/1

Example response (200):

```
{
  "id": 1,
  "operator_id": 1,
  "route_code": "1",
  "route_name": "City Centre - Chapel Allerton",
  "origin": "City Centre",
  "destination": "Chapel Allerton",
  "operator": {
    "id": 1,
    "noc": "YKRT",
    "name": "Example Operator",
    "region": "Yorkshire"
  },
  "stops": [
    {
      "id": 1,
      "stop_code": "45001234",
      "stop_name": "City Square",
      "locality": "Leeds",
      "latitude": 53.796,
      "longitude": -1.548
    }
  ]
}
```

Error: 404 Not Found if route_id does not exist.

3.4 GET /stops

Returns stop records with optional search.

Query parameters:

Name	Type	Description
q	string	Search in stop code/name/locality.
limit	integer (1-100)	Default 20.

Name	Type	Description
offset	integer (≥ 0)	Default 0.

Example request:

GET /stops?q=City&limit=5

Example response (200):

```
[
  {
    "id": 1,
    "stop_code": "45001234",
    "stop_name": "City Square",
    "locality": "Leeds",
    "latitude": 53.796,
    "longitude": -1.548
  }
]
```

3.5 POST /incidents

Creates a new incident. **Requires X-API-Key.**

Request body (JSON):

Field	Type	Required	Constraints
title	string	yes	3-255 chars
description	string	no	max 2000 chars
incident_type	string	yes	2-64 chars
severity	string	yes	low, medium, high
status	string	no	open, monitoring, resolved (default open)
route_id	integer	yes	Must exist
stop_id	integer	no	Must exist if provided

Example request:

POST /incidents
 Content-Type: application/json
 X-API-Key: your-write-api-key

```
{
  "title": "Delay at City Square",
```

```

    "description": "Heavy traffic",
    "incident_type": "congestion",
    "severity": "medium",
    "status": "open",
    "route_id": 1,
    "stop_id": 1
  }

```

Example response (201):

```

{
  "id": 1,
  "title": "Delay at City Square",
  "description": "Heavy traffic",
  "incident_type": "congestion",
  "severity": "medium",
  "status": "open",
  "route_id": 1,
  "stop_id": 1,
  "reported_at": "2025-03-11T12:00:00",
  "updated_at": null
}

```

Errors: 401 Unauthorized (invalid/missing API key), 422 Unprocessable Entity (validation), 404 if route_id or stop_id does not exist.

3.6 GET /incidents

Returns incidents with optional filters.

Query parameters:

Name	Type	Description
route_id	integer	Filter by route.
severity	string	low, medium, high.
status	string	open, monitoring, resolved.
limit	integer (1-100)	Default 20.
offset	integer (≥ 0)	Default 0.

Example request:

GET /incidents?severity=medium&status=open

Example response (200):

```

[
  {

```

```

    "id": 1,
    "title": "Delay at City Square",
    "description": "Heavy traffic",
    "incident_type": "congestion",
    "severity": "medium",
    "status": "open",
    "route_id": 1,
    "stop_id": 1,
    "reported_at": "2025-03-11T12:00:00",
    "updated_at": null
  }
]

```

3.7 GET /incidents/{incident_id}

Returns a single incident.

Path parameters: incident_id (integer).

Example response (200): Same JSON shape as one element in the list above.

Error: 404 Not Found if the incident does not exist.

3.8 PUT /incidents/{incident_id}

Updates an incident. **Requires X-API-Key.** All body fields are optional; only provided fields are updated.

Request body (JSON): Same fields as POST, all optional.

Example request:

```

PUT /incidents/1
Content-Type: application/json
X-API-Key: your-write-api-key

```

```

{
  "status": "resolved",
  "description": "Cleared after 20 minutes"
}

```

Example response (200): Full incident object (same shape as GET /incidents/{id}).

Errors: 401 Unauthorized, 404 Not Found, 422 Unprocessable Entity.

3.9 DELETE /incidents/{incident_id}

Deletes an incident. **Requires X-API-Key.**

Example request:

```
DELETE /incidents/1
X-API-Key: your-write-api-key
```

Response: 204 No Content (no body).

Errors: 401 Unauthorized, 404 Not Found.

3.10 GET /analytics/routes/{route_id}/reliability

Returns route-level reliability metrics for a date range and optional time band.

Path parameters: route_id (integer).

Query parameters:

Name	Type	Description
start_date	string (YYYY-MM-DD)	Start of period.
end_date	string (YYYY-MM-DD)	End of period.
time_band	string	Optional filter (e.g. morning peak).

Example request:

```
GET /analytics/routes/1/reliability?start_date=2025-03-01&end_date=2025-03-10
```

Example response (200):

```
{
  "route_id": 1,
  "start_date": "2025-03-01",
  "end_date": "2025-03-10",
  "time_band": null,
  "avg_delay_minutes": 2.5,
  "on_time_rate": 0.88,
  "cancellation_rate": 0.02,
  "observation_count": 150
}
```

Error: 404 Not Found if the route does not exist or has no metrics.

4. Interactive documentation

When the API is running, OpenAPI (Swagger) UI is available at:

- **Swagger UI:** <http://127.0.0.1:8000/docs>
- **ReDoc:** <http://127.0.0.1:8000/redoc>
- **OpenAPI JSON:** <http://127.0.0.1:8000/openapi.json>

These provide the same endpoints, parameters, and schemas with try-it-out support.